SLEEVEDRIFT · TRACEABILITY MATRIX

SleeveDrift traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sleevedrift		
Source	docs/03b-traceability.md @ 1ba9c34		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-11-02	Amish Chadha	Generated by .kit/trace.py
0.3	2026-11-02	Amish Chadha	Generated by .kit/trace.py
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Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 15 user requirements, 15 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-007	PR-001	Fits the pipe	SVD-CAL-001, SVD-BLD-001, SVD-STM-001	Analysis, SVD-CAL-001 A (TRL 3); fit trial in pipe sections (TRL 4)	Met on paper for 800 mm: every piece passes a 776 mm circle in a 780 mm bore. 600 mm stretch open: held until the co-design partner says whether 600 mm pipes are used for rescue (decision 26C)
UR-001	PR-002	Hood protection	SVD-CAL-001, SVD-BLD-001	Analysis, SVD-CAL-001 B (TRL 3); load test on CalRig (TRL 4)	Met on paper: 5.5 MPa in the crown, tip deflection under 0.1 mm
UR-002	PR-003	Spoil haulage	SVD-CAL-001, SVD-BLD-001, SVD-STM-001, SVD-OPT-001	Analysis, SVD-CAL-001 D (TRL 3); timed trial in a 60 m pipe run (TRL 4)	Met as restated: about 0.42 m3/h with a right-angle drill driving each portal winch (decision 24B), a margin of 5 %; 0.22 m3/h by hand as the fallback, which is below the face rate
UR-003	PR-004	Haul force	SVD-CAL-001, SVD-BLD-001, SVD-OPT-001	Analysis, SVD-CAL-001 C (TRL 3); force gauge during trial (TRL 4)	Met on paper: 93 N at the start in high gear, 30 N in low gear
UR-004	PR-005	Fresh air at the face	SVD-CAL-001, SVD-BLD-001, SVD-STM-001	Analysis, SVD-CAL-001 F (TRL 3); anemometer at duct outlet (TRL 4)	Met on paper: about 15 m3/min at the face
UR-006	PR-006	Casualty evacuation	SVD-CAL-001, SVD-BLD-001	Analysis, SVD-CAL-001 E (TRL 3); timed drill (TRL 4)	Met on paper: about 3.5 min with three people hauling hand over hand
UR-008	PR-007	Rapid set-up	SVD-CAL-001, SVD-BLD-001, SVD-OPT-001	Analysis, SVD-CAL-001 G (TRL 3); timed deployment exercise (TRL 4)	Met on paper: about 1.6 h estimated
UR-009	PR-008	No power in the pipe	SVD-CAL-001	Design review (TRL 3)	Met by design: battery lamps, gas monitors (alarm devices) and sound-powered telephones only; the drill drive stays at the portal
UR-010	PR-009	Shift durability	SVD-CAL-001	Analysis, SVD-CAL-001 H (TRL 3); endurance trial (TRL 4)	Managed: rope abrasion over about 250 round trips at the drill rate cannot be shown on paper; both ropes are inspected against a wear gauge every shift and a spare set is carried (decision 25A). The endurance trial decides whether a tougher rope is needed

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-011	PR-010	Transportable	SVD-CAL-001	Analysis, SVD-CAL-001 I (TRL 3); weighing and handling trial (TRL 4)	Met, thin: heaviest package 37.7 kg (the sledge bundle)
UR-012	PR-011	Overload limit	SVD-CAL-001, SVD-BLD-001, BOM	Analysis, SVD-CAL-001 C and D6 (TRL 3); release test of the swivel pins and proof load on CalRig (TRL 4)	Met by design: rope factor 6.5 on the release load; drill clutch slips at 1.55 kN in high gear
UR-005	PR-012	Air monitoring	SVD-CAL-001, SVD-BLD-001, BOM	Design review (TRL 3); bump test before each shift (TRL 4)	Met by design: two four-gas monitors in the kit
UR-013	PR-013	Affordable to hold	SVD-CAL-001	Priced BOM, bom/bom.csv; SVD-CAL-001 J	Over target: USD 10,804.00 in parts, USD 5,804.00 over. The two winches (USD 2,300), four-gas monitors (USD 1,100), signal line (USD 860), duct (USD 765) and blower (USD 750) are the largest lines. Proposed, awaiting Amish: hold the target, or restate it for a rescue kit
UR-014	PR-014	Made by a general workshop	SVD-CAL-001	Inspection of the BOM, the model and the build plan SVD-BLD-001 (TRL 3)	Met by design: the fidelity check lists every made part (cad/fidelity.json)
UR-015	PR-015	Face and portal signals	SVD-CAL-001	Design review (TRL 3), SVD-DDR-001 item 8; call test (TRL 4)	Met by design: signal line, BOM line 24

Gaps

None.